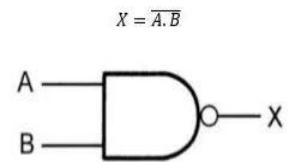
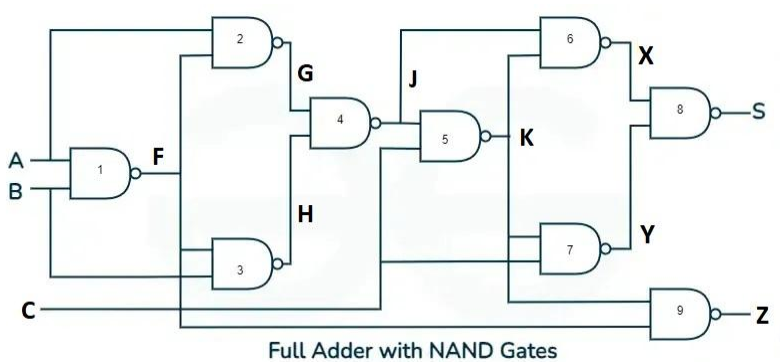


Ken needs to build a Full Adder for his circuit, but all he has currently are blueprints for a two-bit NAND gate. A NAND gate, as shown, takes two inputs (A,B) and has one output (X). Its truth table is also given. A truth table shows what the outputs should be for each input configuration. Two bits have 4 possible input configurations. As can be seen from the truth table, a NAND gate's output is 1 as long as A and B are not both 1.

The Full Adder Ken needs to build adds three bits: two inputs and a carry-in bit (A,B,C). The Full Adder has two outputs: a sum and a carry-out bit (S,Z). A Full Adder is used to add two bits while carrying in from a previous operation and carrying out to the next operation. They can be chained together to add multi-bit values of any size.



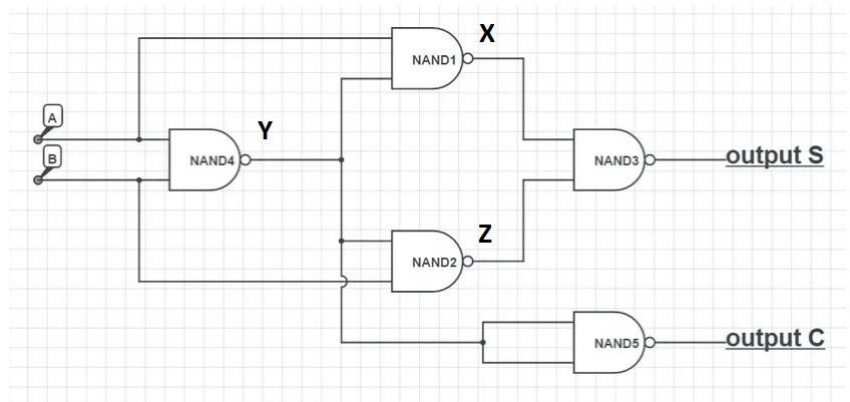
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

NAND gates are considered a universal gate since all digital logic can be derived from them. Logic derived from NAND gates is called NAND logic. Here we have provided the circuit diagram that Ken came up with to implement his full adder. You can find the truth table for this circuit in the student datasets. Help Ken turn his circuit diagram with NAND logic into a program that turns a series of NAND operations into a truth table. This will help him verify that he designed the full adder circuit correctly.

Input

The first 5 lines provide: Title; Inputs; Intermediate Signals; Outputs; and Number (N) of NAND Gates.

If there are no Intermediate Signals, that line will be blank. The last N lines provide the output and the two inputs for each of the NAND Gates. Signals are noted as single capital letters, separated by single spaces. Two examples follow. The first is a single NAND Gate, and the other is a Half Adder. A Half Adder adds only two bits: (A,B). It has two outputs: Sum (S) and Carry (C).



NAND Gate	Half Adder
A B	B A
	Y X Z
X	C S
1	5
X A B	Y A B
	X Y A
	Z Y B
	C Y Y
	S X Z

Output

First print the title. On the next line, print the labels for the Inputs, Intermediate Signals and Outputs in the order provided in the Input file. Separate each letter with a space, and each section with a colon (:). If there are no Intermediate Signals, just print the Outputs. Determine all values by iterating over the Inputs. The Inputs should count up in binary, line by line. On each line, print the values for the Inputs, Intermediate Signals, and Outputs, separated by spaces and colons as in the line of labels. Refer to the examples below.

NAND Gate	Half Adder
A B : X	B A : Y X Z : C S
0 0 : 1	0 0 : 1 1 1 : 0 0
0 1 : 1	0 1 : 1 0 1 : 0 1
1 0 : 1	1 0 : 1 1 0 : 0 1
1 1 : 0	1 1 : 0 1 1 : 1 0

Discussion

There will up to 4 Inputs, and up to 25 NAND Gates (total Intermediate Signal and Outputs). All 26 capital letters may be used. Output signals may be used as inputs to a following NAND Gate. The order of the NAND gates in the Input is the order in which your program should process them.

NAND Background Info

NAND gates are a good way to build up digital circuits using truth tables, which makes them an effective teaching tool. However, transistors are the more authentic building blocks of digital circuits as they model the physical properties of the circuit much more precisely. There are different kinds of transistors but the most common are MOSFET transistors. MOSFET transistors can be thought of as a switch. There are two kinds of MOSFET transistors, NMOS and PMOS. In an NMOS transistor, the switch is closed if you input 1 and is open if you input 0. In a PMOS transistor, the switch is closed if you input 0 and is open if you input 1.

Most MOSFET circuits use a CMOS layout. A CMOS layout is a parallel layout that ensures the output of a circuit is either connected to the ground (bit=0) or connected to the voltage source (bit=1). Using NMOS to connect the ground and PMOS to connect the voltage source is the parallel aspect that ensures you are not connecting your voltage source to the ground (this is called shorting a circuit which is bad). Think of this like when you accidentally breathe in your saliva. Either your mouth should be breathing or swallowing. If it does both, you are doing something wrong. Your larynx is in charge of protecting your lungs from the stuff you were meant to eat. The CMOS layout protects you from connecting your voltage source to ground.

The most simple CMOS circuit is an inverter. Here the parallel layout is very apparent. There is a NMOS on the bottom and a PMOS on top (indicated by the circle on the input). In the case V_{in} is 0, the PMOS switch closes and the NMOS switch opens. The voltage source saturates the V_{out} to be 1. In the case V_{in} is 1, the PMOS switch opens and the NMOS switch closes. The ground saturates the V_{out} to be 0.